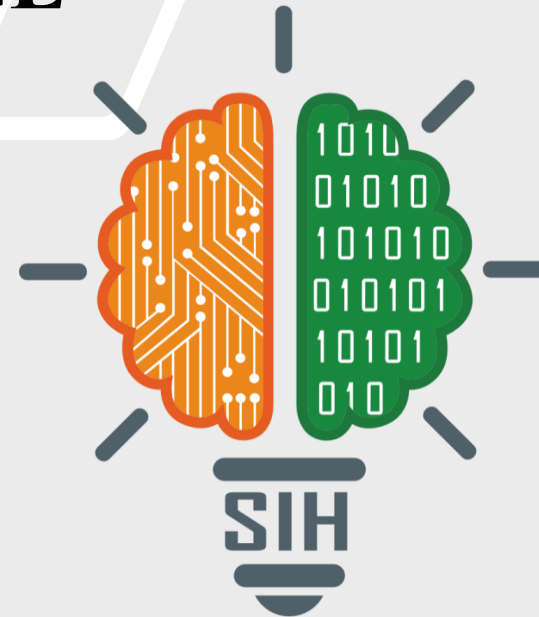


VITISH 2026

(SIH Internal Hackathon)

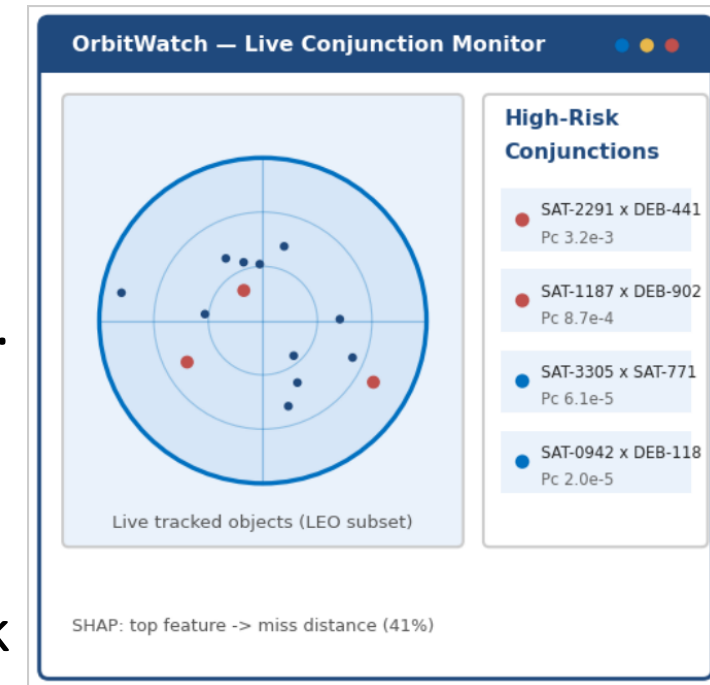
TITLE PAGE

- Problem Statement ID: **17**
- Problem Statement Title: **Space Debris detection & Collision Avoidance**
- Theme: **Space Technology – Space Debris Detection & Collision Avoidance**
- PS Category: **Software**



OrbitGuard: AI-Assisted Space Debris Collision Risk Detection & Avoidance Proposed Solution :

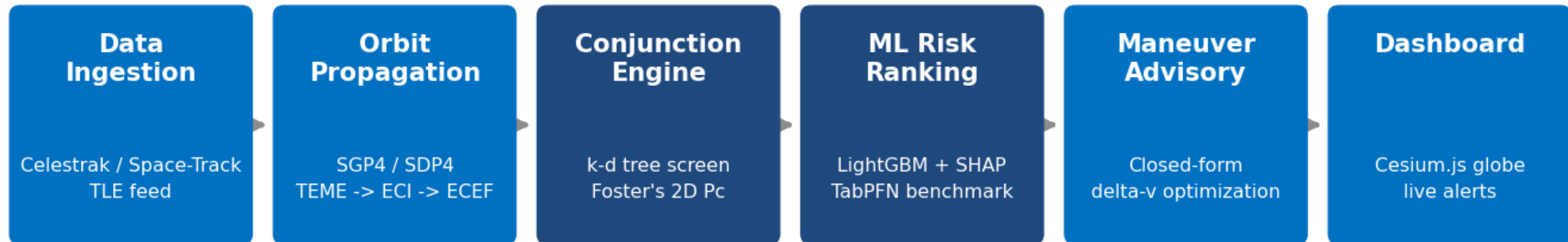
- ❖ Predict satellite/debris trajectories using **TLE + SGP4**.
- ❖ Detect potential close approaches using **conjunction screening**.
- ❖ Calculate **Probability of Collision (Pc)**.
- ❖ Use **LightGBM** to rank collision risks.
- ❖ Use **SHAP** to explain risk factors.
- ❖ Generate **minimum- Δv maneuver advisories** for high-risk events.
- ❖ Display risks through an interactive **3D Cesium.js dashboard**.



TECHNICAL APPROACH

Technologies and Workflow:

- ❖ Python (FastAPI backend)
- ❖ sgp4 + astropy (propagation and frame transforms)
- ❖ scipy k-d tree (conjunction screening)
- ❖ LightGBM + SHAP with a TabPFN benchmark (ML risk ranking)
- ❖ React + Cesium.js (3D globe dashboard)
- ❖ PostgreSQL/SQLite (event storage)
- ❖ Docker Compose (deployment)



SOLUTION FEASIBILITY & VIABILITY

- Uses established **orbital propagation and collision-analysis methods.**
- Uses publicly available **TLE and conjunction datasets.**
- Modular AI + Web architecture enables parallel development.
- Initial demonstration targets **500–1000 LEO objects.**
- **Main risks:** computational load, data integration, ML data availability and live-demo reliability.
- **Mitigation:** staged screening, fixed data schemas, validation and backup demo.

REAL WORLD IMPACTS, BENEFITS & RELEVANCE

- **Improves satellite safety** by identifying dangerous conjunctions early.
- **Prioritizes critical events** using risk ranking.
- Provides **explainable collision-risk analysis**.
- Supports **collision-avoidance decisions** through maneuver advisories.
- Helps protect valuable space infrastructure.
- Supports **safer and more sustainable orbital operations**.

TEAM CAPABILITY

- **Anas – Orbital Mechanics :**
 - TLE,SGP4,coordinate transformations.
- **Rushaan – Collision Analysis :**
 - Conjunction screening, Pc calculation.
- **Udarsh - AI/ML :**
 - LightGBM, SHAP, TabPFN, decision support.
- **Balaganesh - Web Developer :**
 - FastAPI, React, Cesium.js, dashboard.
- **Mahalakshmi - Quality Assurance and Tester :**
 - Pitch, documentation, testing and demo.

RESEARCH AND REFERENCES

- **ESA Kelvins Collision Avoidance Challenge (2019)** - public CDM dataset and ML competition validating gradient-boosted trees, Bayesian NNs and LDA for collision-risk prediction: kelvins.esa.int/collision-avoidance-challenge
- Foster, J.L., "**A Parametric Analysis of Orbital Debris Collision Probability**," NASA - source for the Foster 2D Pc method used in the conjunction engine
- **Celestrak (celestrak.org)** - live TLE data source for tracked LEO objects
- **NASA Orbital Debris Program Office & ESA Space Debris Office** annual environment reports - problem-framing statistics (Kessler syndrome, tracked object counts)
- Hollmann et al., "**Accurate predictions on small data with a tabular foundation model**," Nature 637 (2025) - basis for the TabPFN model-choice benchmark
- **sgp4 (PyPI), astropy, CesiumJS documentation** - core libraries for propagation, coordinate transforms, and 3D visualization